Forecasting 52 weeks ahead

Draft section for paper 1, for review. Planned before it ran (13 November 2026).

1 Forecasting 52 weeks ahead

Over 52 weeks the world model alone is not the most accurate forecast on any data set. The average of the two models, each forecasting all 52 weeks at once, is the most accurate on four of seven data sets, and LightGBM on the other three. Table 1 and Figure 1 give the scores. The test was written down before it ran: the six Dominick's categories and M5, two windows of 52 weeks each (364 days on M5), the tuned settings of the final runs with only the distance ahead changed, scored at every level as in the rest of the paper.

Main comparison, in blocks. Each model forecasts 13 weeks, takes its forecast in as the sales of those weeks, and forecasts the next 13, four times. Here the world model is clearly less accurate than LightGBM in 38 of 72 paired comparisons and clearly more accurate in 5. Against the better of LightGBM's two setups on each data set it is behind on seven of seven. The average in blocks is clearly more accurate than the world model alone in 58 comparisons (clearly less in 2), and than LightGBM in 19 (clearly less in 9).

Taking in its own forecasts costs the world model most. Forecasting all 52 weeks at once, the world model is more accurate than in blocks on six of six data sets; LightGBM is on four of seven. In blocks, the world model's item by store error changes by -0.065 to $+0.237$ from the first 13 weeks to the last, LightGBM's by -0.080 to $+0.150$. The world model rolls its state forward from the sales it takes in; when those are its own forecasts, its errors are carried into the state and grow with each block.

The rule carries to 52 weeks. The paper's rule, the average of the two models, holds at 52 weeks when each model forecasts the whole year at once. In paired tests the average forecasting directly is clearly more accurate than LightGBM forecasting directly in 15 of 48 comparisons and clearly less in 2; against the world model forecasting directly it is clearly more accurate in 37 and clearly less in 0. These two comparisons were added after the planned ones.

Limits. Two windows per data set and two retailers (six Dominick's categories and M5). The plan is the one recorded, as everywhere in the paper. The settings were tuned for 8 weeks (28 days on M5) and not tuned again for 52. M5's world model was not run forecasting all 52 weeks at once, because rolling its state 364 days in training is beyond the compute of this test.

Table 1: Forecasts 52 weeks ahead (364 days on M5). WRMSSE over the 52 weeks, mean over levels and two windows, for the world model (WM), LightGBM (LGB) and their average (Avg). In blocks: each model forecasts four blocks of 13 weeks, each block's forecast taken in as sales before the next. Direct: each model forecasts all 52 weeks at once (not run for M5's world model). Paired item bootstrap over levels and windows: comparisons clearly lower / clearly higher, of all. Last four columns: WRMSSE item by store in the first and last block of 13 weeks, in blocks. Shading marks the lowest of the six: world model, LightGBM, average.

	In blocks			Direct			Paired		WM, item by store		LGB, item by store	
Data set	WM	LGB	Avg	WM	LGB	Avg	WM vs LGB, blocks	Avg vs LGB, direct	Weeks 1 to 13	40 to 52	Weeks 1 to 13	40 to 52
Analgesics	1.062	**0.865**	0.907	0.964	0.867	0.868	0 / 3 of 8	1 / 0 of 8	0.697	0.686	0.696	0.683
Bottled juice	0.903	0.797	0.817	0.764	0.769	**0.726**	0 / 2 of 8	4 / 0 of 8	0.738	0.742	0.705	0.689
Cheese	1.086	0.689	0.820	0.773	0.707	**0.677**	0 / 8 of 8	2 / 0 of 8	0.701	0.938	0.692	0.738
Frozen entrees	0.424	0.397	0.393	0.412	0.393	**0.387**	0 / 5 of 8	2 / 0 of 8	0.469	0.404	0.450	0.370
Soft drinks	0.872	0.862	0.819	0.851	**0.754**	0.767	1 / 0 of 8	0 / 2 of 8	0.618	0.767	0.631	0.781
Snacks	1.013	0.877	0.920	0.866	0.882	**0.843**	0 / 6 of 8	6 / 0 of 8	0.691	0.890	0.622	0.683
M5	1.016	0.931	0.874	n/a	**0.839**	n/a	4 / 14 of 24	n/a	0.823	1.000	0.849	0.908

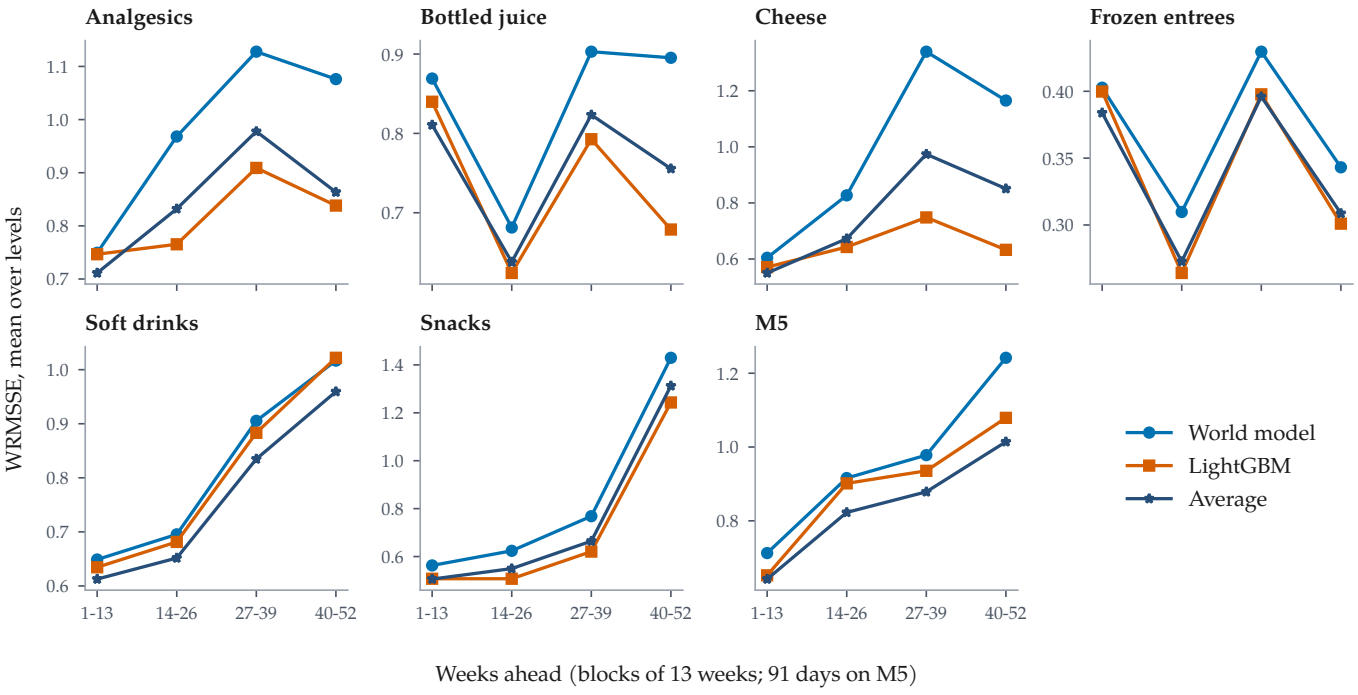

Figure 1: WRMSSE, mean over levels and two windows, in each block of 13 weeks (91 days on M5), for the world model, LightGBM and their average forecasting in blocks.